



Figure 2: Four configurations of an e-learning setup as applied to virtual classrooms using WebEx

Launching mandatory new e-learning courses developed in-house becomes easier. The next logical step for any enterprise is setting up an in-house e-learning content development team managed by the competency development function. The content team takes the responsibility of creating custom content relevant to the IT business by working with SMEs (subject matter experts). Such teams in any large scale enterprise can create, on an average, 800 to 1,000 e-learning courses annually, using instructional design techniques of varying complexity that eventually get ported on the LMS for employees to learn and stay relevant in a competitive IT business. This makes the company self-reliant and its e-learning goals sustainable.

To summarise, at this stage, the following are carried out successfully in the journey of adopting e-learning.

- Transitioning from vendor based hosting of content to having in-house hardware and software (a rack of servers) to deliver e-learning courses.
- An in-house LMS infrastructure (customised to needs) to launch new courses in a manner that allows management to promote a course, measure the progress made by an employee on a given course, assign courses to a set of employees (like mandatory courses for compliance that need to be assigned to some people within the company) who need to complete them in a given period of time, take feedback from employees on the courses, provide course previews and user ratings on the courses, etc.
- The creation of a platform for launching in-house course content for users across the enterprise.
- The setting up of a team of experts developing in-house e-learning content for the enterprise.

E-learning through virtual classroom technologies

Trying to scale e-learning programmes in an IT company with a highly distributed workforce is one more significant challenge for the competency function. And if the scaling needs to be achieved with a limited increase in training staff,

only a technological solution can come to the rescue. This challenge motivated the competency functions to explore the option of whether classrooms could be made virtual.

Virtual classrooms serve as the next-generation e-learning environment in many companies. The simplest of these are the video conferencing facilities used for training. This is the most basic and simplest of the solutions, transforming a set of meeting rooms in any enterprise that are connected by a video conferencing facility, into a classroom. Here, the connected participants experience learning by following the instructor located in a remote location, virtually. There are limitations to how much learning engagement can happen through this medium. This is Level-1 in virtual training wherein the learning content and audio can be synced across locations, but the collaborative aspects of learning remain very poor. The participant can often go offline from learning, get demotivated and, for the faculty, it remains a challenge to maintain contact with all the participants. Long sessions become ineffective very quickly.

With the rapid advancement of Web related technologies, the software's capability to virtualise the learning environment has only become better. Two technologies that are ideal at this stage of the evolution of e-learning are:

- CISCO WebEx
- Microsoft Lync

Do keep in mind that there are many other similar related technologies and software solutions.

The authors have experienced these two technologies that enable virtual learning and can help scale a classroom session reasonably well.

The WebEx training centre has a rich UI that helps the instructor to explain concepts and interact with participants. There is the white board, chat centre, pointer, polling, voice management centre, etc. Lync is a lean solution which has much higher scalability in an enterprise that has Windows serving as the backbone operating system for the enterprise computing environment. There are other virtual classroom environments that are highly scalable, such as Monarch (<http://www.monarchinnovative.com/>) which converts the faculty seat into a studio with a wide screen in it. The remote classrooms are projected on the wide screen in front of the faculty, all at the same time. The faculty can view all the participants and interact with them to improve the learning environment. This solution, however, can only scale an ILT classroom by eight to ten times, not more.

Regarding business benefits, a virtual e-learning architecture for classrooms enables the following:

- Reduces or eliminates the need for faculty to travel to multiple locations, resulting in cost savings.
- Increased programme reach — for example, the same sessions are available to many employees at the same time.
- Less time is needed to schedule a programme and the